TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	58.9 ± 0.30
Correlation Stability	69.8 ± 0.40
Distribution Consistency	73.8 ± 0.98
Entropy Preservation	32.3 ± 1.10
Mean Z Score Drift	0.5 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	29.6 ± 5.44
Data Density	100.0 ± 0.00
Coverage Score	31.3 ± 6.13
Feature Accuracy	38.6 ± 0.49
Novelty Score	0.6 ± 0.05

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Dataset Audit: chronic_kidney_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	400
Columns	25
Numeric features	25
Categorical features	0
Missing cells (%)	10.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	51.483	45.772	-5.711	11.1%	Warn
bp	76.469	72.445	-4.024	5.3%	OK
sg	1.017	1.022	+0.005	0.4%	OK
al	1.017	0.000	-1.017	100.0%	Drift
su	0.450	0.000	-0.450	100.0%	Drift
rbc	0.190	0.000	-0.190	100.0%	Drift
pc	0.227	0.000	-0.227	100.0%	Drift
pcc	0.106	0.000	-0.106	100.0%	Drift
ba	0.056	0.000	-0.056	100.0%	Drift
bgr	148.037	105.335	-42.702	28.8%	Drift
bu	57.426	32.506	-24.920	43.4%	Drift
sc	3.072	0.904	-2.168	70.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	58.9 ± 0.30
Correlation Stability (*2)	69.8 ± 0.40
Distribution Consistency (*3)	73.8 ± 0.98
Entropy Preservation (*4)	32.3 ± 1.10
Mean Z Score Drift (*5)	0.5 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	29.6 ± 5.44

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	31.3 ± 6.13
Feature Accuracy (*10)	38.6 ± 0.49
Novelty Score (*11)	0.6 ± 0.05

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9574999999999999	N/A	0.95	-0.007500000000000007	
F1	0.9607047320487891	N/A	0.956171071525179	-0.004533660523610117	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9525	N/A	0.95625	0.003750000000000009	
F1	0.9587663012688215	N/A	0.9622501147783022	0.003483813509480682	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6024999999999999	N/A	0.53375	-0.06874999999999999	
F1	0.7507214882964476	N/A	0.4917752249742129 4	-0.2589462633222348	*

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9100000000000001	N/A	0.9200000000000002	0.010000000000000009	
F1	0.92384446063128	N/A	0.9333320302506314	0.0094875696193514	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9887500000000001	N/A	0.9825000000000002	-0.00625	
F1	0.990128320402674	N/A	0.9849125322435064	-0.005215788159167515	

Model: MLP

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7725	N/A	0.7137500000000001	-0.05875000000000001	
F1	0.7755951540418404	N/A	0.7268144270658973	-0.04878072697594309	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6125	N/A	0.60375	-0.00874999999999997	
F1	0.5885198185573248	N/A	0.5246618721364005	-0.06385794642092436	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.97875	N/A	0.9824999999999999	0.00374999999999987	
F1	0.9805471229310466	N/A	0.9849841858176441	0.004437062886597631 5	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9825000000000002	N/A	0.9912500000000002	0.008750000000000003	
F1	0.9848619694663338	N/A	0.9926382008307314	0.007776231364397557	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9974999999999999	N/A	0.9962500000000001	-0.001249999999999995 5	
F1	0.9978258300838945	N/A	0.9968837023376433	-0.000942127746251453 3	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	48.0	80.0	1.02	1.0	0.0	nan	0.0	0.0	0.0	121.0
2	7.0	50.0	1.02	4.0	0.0	nan	0.0	0.0	0.0	nan
3	62.0	80.0	1.01	2.0	3.0	0.0	0.0	0.0	0.0	423.0
4	48.0	70.0	1.005	4.0	0.0	0.0	1.0	1.0	0.0	117.0
5	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
6	60.0	90.0	1.015	3.0	0.0	nan	nan	0.0	0.0	74.0
7	68.0	70.0	1.01	0.0	0.0	nan	0.0	0.0	0.0	100.0
8	24.0	nan	1.015	2.0	4.0	0.0	1.0	0.0	0.0	410.0
9	52.0	100.0	1.015	3.0	0.0	0.0	1.0	1.0	0.0	138.0
10	53.0	90.0	1.02	2.0	0.0	1.0	1.0	1.0	0.0	70.0
11	50.0	60.0	1.01	2.0	4.0	nan	1.0	1.0	0.0	490.0
12	63.0	70.0	1.01	3.0	0.0	1.0	1.0	1.0	0.0	380.0
13	68.0	70.0	1.015	3.0	1.0	nan	0.0	1.0	0.0	208.0
14	68.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	98.0
15	68.0	80.0	1.01	3.0	2.0	0.0	1.0	1.0	1.0	157.0
16	40.0	80.0	1.015	3.0	0.0	nan	0.0	0.0	0.0	76.0
17	47.0	70.0	1.015	2.0	0.0	nan	0.0	0.0	0.0	99.0
18	47.0	80.0	nan	nan	nan	nan	nan	0.0	0.0	114.0
19	60.0	100.0	1.025	0.0	3.0	nan	0.0	0.0	0.0	263.0
20	62.0	60.0	1.015	1.0	0.0	nan	1.0	1.0	0.0	100.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	36.27067910727921	60.6077674106006	1.02	0.0	0.0	0.0	0.0	0.0	0.0	107.72368843773862
2	35.45880437763837	62.66061875202952	1.025	0.0	0.0	0.0	0.0	0.0	0.0	105.8578971904428
3	55.25336614412556	79.15544618624816	1.025	0.0	0.0	0.0	0.0	0.0	0.0	126.45287398224224
4	46.48778556753544	65.3243465670062	1.02	0.0	0.0	0.0	0.0	0.0	0.0	109.53917473144597
5	50.81380683206947	65.75575284669561	1.02287787642334	0.0	0.0	0.0	0.0	0.0	0.0	115.33132813136518
6	50.39538466178561	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	78.30345214741291
7	35.0	80.0	1.005	3.0	0.0	1.0	0.0	0.0	0.0	nan
8	30.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.0
9	52.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	99.0
10	65.0	80.0	1.015	0.0	0.0	nan	0.0	0.0	0.0	115.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	42.241618297446216	70.77601143590388	1.02038800571795 2	0.0	0.0	0.0	0.0	0.0	0.0	124.53439313845767
2	51.40545267706214 4	63.18909464587571	1.025	0.0	0.0	0.0	0.0	0.0	0.0	118.232366252113
3	35.27917625407112	64.55835250814224	1.025	0.0	0.0	0.0	0.0	0.0	0.0	91.3958812703556
4	35.6526714312278	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	93.28831428939235
5	68.40344016242034	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	109.76440857722923
6	64.42384718367725	73.94038204080687	1.02197019102040 33	0.0	0.0	0.0	0.0	0.0	0.0	87.69538873470901
7	64.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	113.0
8	42.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	98.0
9	12.0	60.0	1.015	3.0	0.0	1.0	1.0	1.0	0.0	nan
10	23.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.0

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	57.110837677955374	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	111.85549762726168
2	25.663644493126824	78.59808184572272	1.0242990409228614	0.0	0.0	0.0	0.0	0.0	0.0	108.4765473223009
3	48.96737093905239	78.10460683690775	1.0204738482907731	0.0	0.0	0.0	0.0	0.0	0.0	127.2418427347631
4	61.54118421483989	65.08236842967978	1.02	0.0	0.0	0.0	0.0	0.0	0.0	104.86821051251236
5	41.81557406457298	69.90778703228649	1.025	0.0	0.0	0.0	0.0	0.0	0.0	75.20286852896972
6	34.95682995050637	61.20186473499625	1.0243990676325019	0.0	0.0	0.0	0.0	0.0	0.0	112.44223768199551
7	34.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	121.0
8	33.0	60.0	nan	nan	nan	0.0	0.0	0.0	0.0	130.0
9	60.0	70.0	nan	nan	nan	nan	nan	0.0	0.0	124.0
10	57.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	132.0

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	71.87051087786055	60.86883778626111	1.02	0.0	0.0	0.0	0.0	0.0	0.0	126.65246488549556
2	59.5399005083931	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	129.87605969496414
3	40.09087884003028	73.93919226686856	1.02	0.0	0.0	0.0	0.0	0.0	0.0	112.60574801364626
4	29.370920459417001	73.62907954058299	1.025	0.0	0.0	0.0	0.0	0.0	0.0	110.04436193108745
5	52.945137067555443	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	108.43068246995873
6	52.23085845648982	75.02251946295316	1.0237556298657382	0.0	0.0	0.0	0.0	0.0	0.0	118.76463765091955
7	44.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0
8	55.0	90.0	1.01	2.0	1.0	1.0	1.0	0.0	0.0	273.0
9	45.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	360.0
10	72.0	90.0	nan	nan	nan	nan	nan	0.0	0.0	308.0

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	59.880195417638646	66.86634097060227	1.02	0.0	0.0	0.0	0.0	0.0	0.0	103.2534636117591
2	40.094933561488276	73.9662237432551	1.02	0.0	0.0	0.0	0.0	0.0	0.0	112.6476468020455
3	40.756890712572044	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	100.05835519845101
4	38.80086050997711	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	96.85959900161791
5	31.740524180139126	80.0	1.0240701570021127	0.0	0.0	0.0	0.0	0.0	0.0	70.74387439830971
6	55.0	70.70750404062053	1.02	0.0	0.0	0.0	0.0	0.0	0.0	109.33476333404778
7	41.0	80.0	1.015	1.0	4.0	1.0	0.0	0.0	0.0	210.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	87.0
9	26.0	70.0	1.015	0.0	4.0	nan	0.0	0.0	0.0	250.0
10	50.0	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	137.0

B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	72.10156401804424	61.2834799742225	1.0206417399871113	0.0	0.0	0.0	0.0	0.0	0.0	124.433040051555
2	38.55295564665432	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	86.13747729367057
3	56.60062019839396	70.82316458918002	1.0245884177054099	0.0	0.0	0.0	0.0	0.0	0.0	90.79875960321208
4	41.20249664874039	77.9750335125961	1.025	0.0	0.0	0.0	0.0	0.0	0.0	111.59500670251921
5	48.56825159626667	65.68251596266671	1.02	0.0	0.0	0.0	0.0	0.0	0.0	119.10314495333338
6	41.75432236840405	72.45677631595947	1.0212283881579798	0.0	0.0	0.0	0.0	0.0	0.0	97.667875000323
7	8.0	60.0	1.025	3.0	0.0	0.0	0.0	0.0	0.0	78.0
8	60.0	70.0	1.01	1.0	0.0	nan	0.0	0.0	0.0	109.0
9	nan	70.0	1.01	3.0	0.0	0.0	0.0	0.0	0.0	110.0
10	62.0	90.0	1.02	2.0	1.0	nan	0.0	0.0	0.0	169.0

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	47.637393901066304	80.0	1.0187631936457575	0.0	0.0	0.0	0.0	0.0	0.0	98.86729136939482
2	66.08461534709014	70.74753452638714	1.0203737672631936	0.0	0.0	0.0	0.0	0.0	0.0	82.92524654736128
3	20.385563747939575	62.57042498626383	1.0243573937534344	0.0	0.0	0.0	0.0	0.0	0.0	118.1725325151098
4	48.00659921092897	74.96700394535517	1.0178656209162433	0.0	0.0	0.0	0.0	0.0	0.0	102.05939289836068
5	35.101735176076716	72.58179887815737	1.025	0.0	0.0	0.0	0.0	0.0	0.0	129.41462459955477
6	62.34543389672609	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	99.01425438659432
7	nan	70.0	1.02	0.0	0.0	nan	nan	0.0	0.0	219.0
8	29.0	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	127.0
9	50.0	90.0	1.015	1.0	0.0	1.0	1.0	0.0	0.0	nan
10	65.0	70.0	1.02	1.0	0.0	1.0	1.0	0.0	0.0	139.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	52.662496020587206	70.0	1.0205081530390028	0.0	0.0	0.0	0.0	0.0	0.0	106.2885857453961
2	57.793201479160896	64.55406845546176	1.025	0.0	0.0	0.0	0.0	0.0	0.0	100.05128510487589
3	34.225975879192276	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	99.12866733712433
4	50.509613321746706	62.193508326091689	1.02	0.0	0.0	0.0	0.0	0.0	0.0	131.73558001737993
5	38.9820519529178	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	115.50448701177055
6	37.059178342310695	80.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.35502783717871
7	53.0	100.0	1.01	1.0	3.0	1.0	0.0	0.0	0.0	213.0
8	34.0	70.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	nan
9	45.0	80.0	1.02	3.0	0.0	0.0	1.0	0.0	0.0	425.0
10	33.0	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	89.0

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	45.6948981631767 2	62.1751697280388 1	1.02	0.0	0.0	0.0	0.0	0.0	0.0	115.042347244765 07
2	60.9040096816509 5	69.9400060510318 4	1.02497000302551 58	0.0	0.0	0.0	0.0	0.0	0.0	132.886011496960 5
3	46.3665136115252 8	80.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	127.474701094401 6
4	25.0579128685251 08	60.1286952633891 3	1.02003217381584 73	0.0	0.0	0.0	0.0	0.0	0.0	119.012869526338 92
5	57.5606258350915 7	73.8215268498982 6	1.01914368530024 91	0.0	0.0	0.0	0.0	0.0	0.0	125.528610739959 3
6	49.5593648049283 8	74.6582694177650 4	1.02232913470888 25	0.0	0.0	0.0	0.0	0.0	0.0	111.440635195071 62
7	46.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	102.0
8	22.0	60.0	1.025	0.0	0.0	0.0	0.0	0.0	0.0	97.0
9	65.0	80.0	1.01	3.0	3.0	nan	0.0	0.0	0.0	294.0
10	23.0	60.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	95.0

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	bp	sg	al	su	rbc	pc	pcc	ba	bgr
1	37.5991596067409 9	72.2494747542131 3	1.02387526262289 33	0.0	0.0	0.0	0.0	0.0	0.0	115.601260589888 51
2	47.6935700183393 6	70.0	1.02	0.0	0.0	0.0	0.0	0.0	0.0	124.313479998119 04
3	61.0273205933803 6	72.3744474777665 8	1.02	0.0	0.0	0.0	0.0	0.0	0.0	81.4748894955533 1
4	54.0	71.987912322196	1.01943758352450 33	0.0	0.0	0.0	0.0	0.0	0.0	80.7709895732703 8
5	35.7370138086194 1	66.8685069043097	1.025	0.0	0.0	0.0	0.0	0.0	0.0	81.8892848105186 5
6	47.3117141908762 3	80.0	1.02277450215013 27	0.0	0.0	0.0	0.0	0.0	0.0	74.4392034402124 2
7	11.0	80.0	1.01	3.0	0.0	nan	0.0	0.0	0.0	nan
8	61.0	90.0	1.015	0.0	2.0	nan	0.0	0.0	0.0	nan
9	51.0	80.0	1.01	2.0	0.0	0.0	0.0	0.0	0.0	106.0
10	70.0	60.0	nan	nan	nan	nan	nan	0.0	0.0	120.0